

Topic: Architecture of 8086 Family

Q. Describe the internal architecture of the 8086 Microprocessor. Explain the functions of the Bus Interface Unit (BIU), Execution Unit (EU), Registers, and Memory Segmentation.

📌 Definition to Remember

The **Intel 8086** is a 16-bit microprocessor with a 16-bit data bus and a 20-bit address bus (can address 1 MB of memory). Its internal architecture is divided into two parallel functional units: the **Bus Interface Unit (BIU)** for memory/bus operations, and the **Execution Unit (EU)** for instruction execution, enabling early pipelining.

1. Bus Interface Unit (BIU)

Handles all data and address transfers on the system buses. The EU relies entirely on the BIU to fetch data.

* **Functions:** Fetches instructions from memory, reads/writes data to memory/IO, calculates physical addresses.

* **Instruction Queue:** A 6-byte pipeline queue. BIU pre-fetches up to 6 bytes of instructions ahead of time to keep the EU busy (Instruction Pipelining).

2. Execution Unit (EU)

Responsible for decoding and executing instructions.

* **Functions:** Fetches instructions from the BIU's queue, decodes them, passes operands to the ALU, and executes the operation.

* It does NOT connect directly to the system buses.

3. Registers of 8086 (16-bit)

Category	Registers
General Purpose	AX (Accumulator), BX (Base), CX (Count), DX (Data). Can split into 8-bit (AH , AL).
Pointer & Index	SP (Stack Pointer), BP (Base Pointer), SI (Source), DI (Destination).
Segment Registers	CS (Code Segment), DS (Data Segment), SS (Stack Segment), ES (Extra Segment).
Special	IP (Instruction Pointer), Flag Register (Status flags like Zero, Carry).

4. Memory Segmentation

The 8086 has 16-bit registers but needs to access 20-bit addresses (1 MB). Memory is logically divided into segments of **64 KB** each.

* **Physical Address Calculation:** The 20-bit address is created by shifting a Segment Register left by 4 bits (multiplying by 16 or 10H) and adding a 16-bit Offset.

* **Formula:** $\text{Physical Address} = (\text{Segment Register} * 10H) + \text{Offset}$

* **Example:** If CS = 1000H and IP = 0002H , Physical Address = 10000H + 0002H = 10002H .

★ Must-Write Points (for 10 marks)

1. 8086 is a 16-bit processor; 16-bit data bus; 20-bit address bus (1 MB memory).
2. Divided into two units: **Bus Interface Unit (BIU)** and **Execution Unit (EU)**.
3. **BIU**: Handles bus operations, calculates addresses, manages a 6-byte Instruction Queue (Pipelining).
4. **EU**: Decodes and executes instructions fetched from the BIU queue using the ALU.
5. Registers: General Purpose (AX, BX, CX, DX), Segment (CS, DS, SS, ES), Pointers (SP, BP, IP), Flags.
6. **Memory Segmentation**: Divides 1 MB memory into 64 KB logical segments.
7. **Physical Address Calculation**: $(\text{Segment} * 10\text{H}) + \text{Offset}$ (solves 20-bit address using 16-bit registers).

⚡ Quick Recall

8086 → 16-bit Data, 20-bit Address (1MB) → BIU (Bus/Memory, 6-byte Queue, Address Calc) + EU (Decode/Execute, ALU) → Registers (AX, BX, Segments, Pointers) → Segmentation (Physical Addr = Segment*10H + Offset)